

TITLE : HEMOPULSE AI PRO - AI-POWERED BLOOD BANK MANAGEMENT PLATFORM

ABSTRACT

HemoPulse AI Pro is an Artificial Intelligence-powered Blood Bank Management System developed to improve the efficiency of blood bank operations through intelligent inventory management and machine learning-based demand forecasting. The system assists hospitals and blood banks in predicting future blood demand, monitoring blood inventory, managing donors, processing blood requests, and identifying possible shortages before they become critical. Existing blood bank management systems mainly focus on maintaining donor records and blood inventory while relying on manual estimation for future demand. Such systems often lead to overstocking, blood wastage, or unexpected shortages due to the absence of predictive analytics and intelligent decision support. They also provide limited security features and lack role-based access control, automated notifications, and staff approval mechanisms.

The proposed system overcomes these limitations by integrating LightGBM and Meta Prophet machine learning algorithms to forecast blood demand several days in advance and generate intelligent shortage alerts. The application is developed using the Flask framework and provides secure user authentication with email OTP verification, password hashing, role-based access control, staff approval by administrators, donor management, blood request management, inventory monitoring, emergency notifications, and interactive dashboards for different categories of users. The system consists of three types of users: Administrator, Staff, and Public User. The Administrator manages users, approves staff registrations, monitors inventory, supervises AI predictions, and generates reports. Staff members maintain blood inventory, process blood requests, organize donation camps, and monitor shortages, while Public Users can securely register, verify their accounts, search blood availability, register as donors, submit blood requests, and track their request status through a personalized dashboard.

The application is implemented using Python, Flask, HTML5, CSS3, Bootstrap 5, JavaScript, SQLite3, SQLAlchemy, Flask-Login, Flask-Bcrypt, Flask-Mail, Pandas, NumPy, Scikit-learn, LightGBM, Meta Prophet, Plotly, Chart.js, Joblib, Git, and GitHub. By combining intelligent forecasting with secure blood bank management, HemoPulse AI Pro helps healthcare organizations improve blood availability, reduce wastage, enhance operational efficiency, and support faster decision-making during emergency situations.

Keywords: Artificial Intelligence, Blood Bank Management, Machine Learning, Blood Demand Forecasting, LightGBM, Meta Prophet, Flask, Blood Inventory Management, Predictive Analytics, Email OTP Authentication, Healthcare Analytics.